1A - Intro to Psychology

module - 1
introduct to psychology
- what is psycology
- History of psycology
- Scope of psydology
- nature Vs nurture
- Scientific psycology and methods.

module - 2
Biology and Behaviour.
- species typical Behaviour pattern
- Brain and Behaviour
- neurons and synapase
- ~~number system~~ nervous system and its functions and Behaviour
- Hemisphere

module - 3
Emotions
- Expression and perception of emotions
- physiology of emotions
- theories of emotions

module - 4
learning
- defination and perspectives of learning
- classical conditioning
- instrumental conditioning
- Skinner reinforcement
- sceduls of reenforcement

cognitive learning

Psychology :

Study of the mind and human and animal behavior.

→ Cicarelli & Mistra ; Atkinson & Hilgard : psychology can be defined as the scientific study of behavior and mental processes.

→ Morgan & King : science of human and animal behaviour ; it includes the application of this science to human problems.

End sem - 50 %

mid sem = 40 %

~~Attendance~~ & class activity = 10 %

Behavior : anything that a person or animal does that can be observed in some way.

→ can be recorded and studied - see and hear.

→ shrink - therapist.

Scope of psychology.

(1) Brain Damage and face recognition → distinct → forgotten.

→ the man who mistook his wife for a hat" - prosopagnosia.

(2) Attributing traits to people :

(3) childhood amnesia -
→ almost no one can recall events from the first 3 years of life

(4) Obesity - increase -

(5) ~~media~~ V. ~~Effects~~ of media violence on children's agression.
→ watching violence of TV causes children to be more aggressive.
→ cathartic

1.1 Bit of History :-
- Socrates, plato, and Arustotle, posed fundamental questions about mental life.

- what is consciousness ?
- Are people inherently rational or irrational ?
- Is there really such a thing as free choice ?

Nature - Nurture debate.
→ the queton of wheather human capabilities are inborn or acquired through experience.

John - locke ⟶ gave concept of tabula rasa.

→ father of psychology - Wilhelm Wundt established the first psychological laboratory at the university of Leipzig in Germany in 1879.

INTROSPECTION : refers to observing and recording the nature of one's own perception, thoughts, and feelings.

Analogy : It's like being asked to describe step-by-step, the exact taste, texture, and smell of a new food you're trying.

School of thought :- Structuralism.
→ championed in the U.S by E.B. Titchener, who was trained by Wundt.

School of thought : Behaviourism.
→ John-B-watson, who reacted against studying conscious experience.

The Rule :- for psychology to be a true science, it must focus only on what is pubic and observable - behaviour

The mind is a 'black box' we can't see inside, so we shouldn't guess, we can only study the stimulus (input) and the response (output)
S-R

core belief: nearly all behavior is a result of conditioning, where the environment shapes us by reinforcing certain habits.

B F Skinner: for example, giving children cookies to stop them from whining reinforces (rewards) the habit of whining.

School of thought: Gestalt psychology.
→ Appeared in Germany around 1912:

∧ - our brain completes smtg that is
< > incomplete.

primary interest: perception.

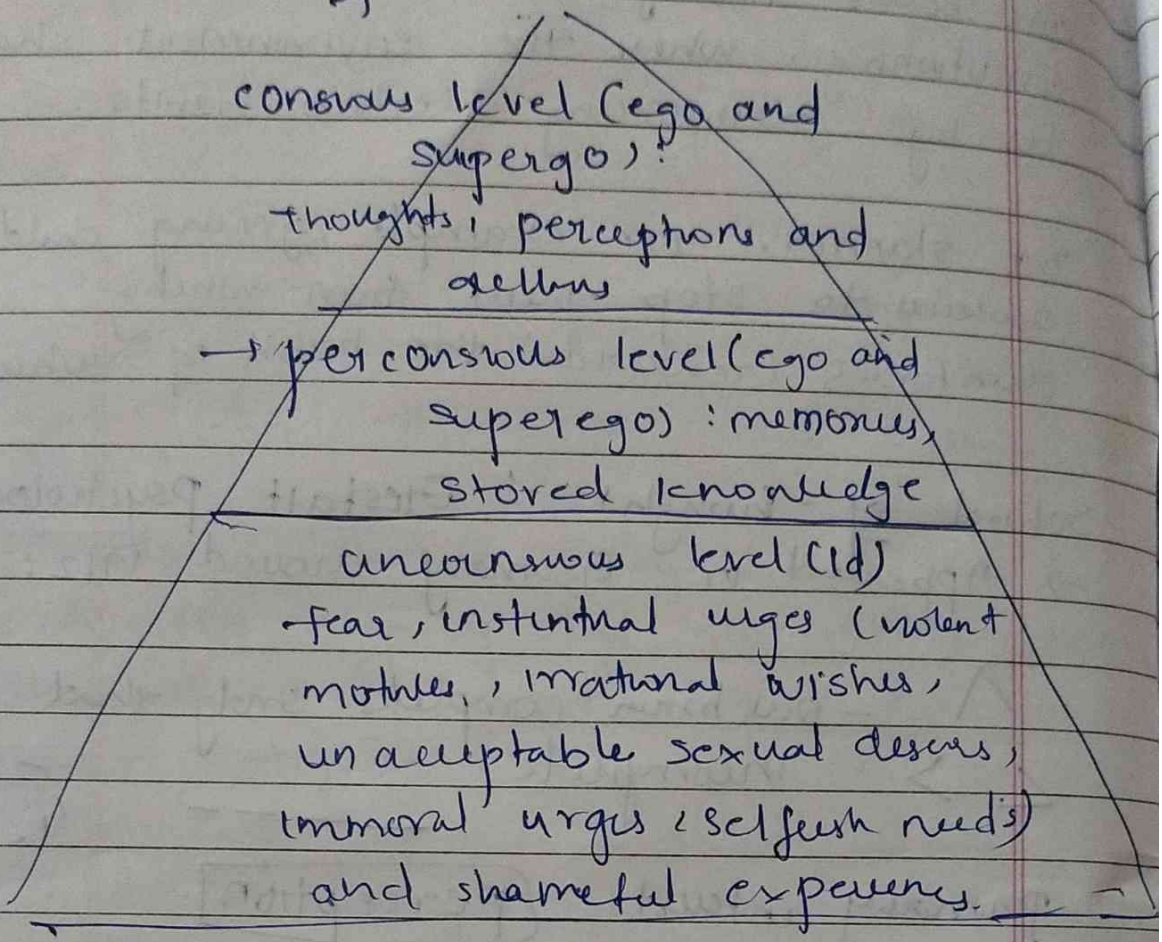
Central idea: the whole is different from the sum of its part.

perception: understanding.

→ our minds automatically organize stimuli into meaningful patterns. we perceive a unified whole, not just isolated bits of information.

School of thought: psychoanalysis.
→ Sigmund Freud: unconscious mind shapes behaviour.
Personality: id (instincts), Ego (reality), superego (morals)

Structur of the mind



core concept: the unconsous - the part of our mind holding thoughts, wishes, and impulses that we are unaware of

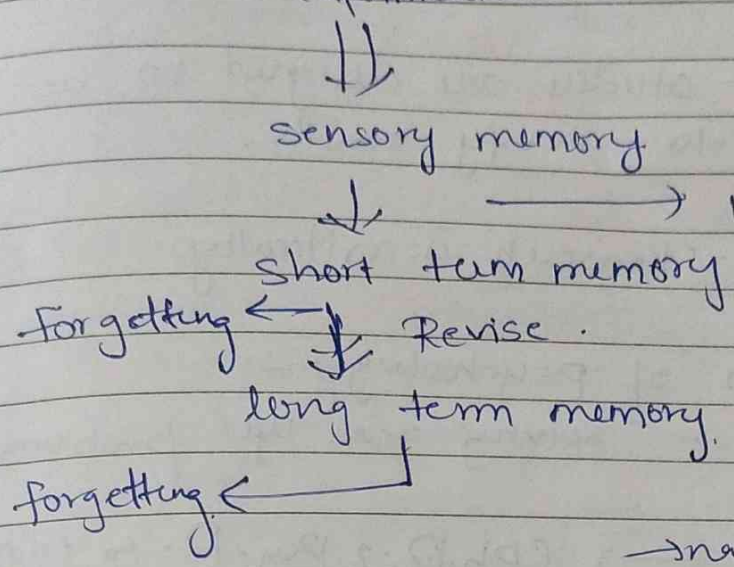
later developments: The cognitive Revolution.

→ The mind Returns.

→ After WWII, it became clear that earlier theories like behaviorism were too restrictive.

① Information-processing:
② Psycholenguistics:
③ Neuropsychology.

①



→ nature nurture worked together

Psychology as a science:

science + → body of systematized knowledge that is gathered by carefully observing and measuring events. (morgan & king 1963)

→ Observation & measurement.

* systematic observation - psychologists observe and measure behaviour under controlled conditions.

* Testable theories - organization of observed facts - psychological theories are falsified and refined based on findings.

* measurement - assignment of numbers to objects or events according to certain rules.

* Empirical Evidence : Data is gathered thro Experiments, surveys, and observations.

* Replication - studies are designed to be repeated to verify results.

* peer Review - Research is critically.

The application of psychology :
- Applied side : "solving real life problems"

psychologist → (Ph.D. / Psy. D. or Ed.D)
psychiatrist → medical doctor.
psychiatric social worker - has training in area of social work.

Methods in psychology.
→ steps in the scientific approach.

* watching cartoon makes children angry (Observation)

* why & HOW (Research)

→ (Theory, Theory. the world is round) (then a person should be able to sail in a straight line around the world and come back to where he or she started) → Hypothesis
→ TENABLE
→ CONFIRMATION BASIS
→ REPEATED EXPERIMENT

Observation
↓
Research question/problem
↓
Hypothesis Generation

- Hypothesis Testing
  ↓
  Drawing conclusion
  ↓
  Reporting results.

→ Descriptive method:
Naturalistic observation.
- participant → Be part of the Group.
- non participant → stands far and observes.
  ↓
  may change if they you know they observe

Disadvantages:
① observer bias:
② Lack of control:

Descriptive method: case study.
case study: researches try to learn everything they can about that individual

- Advantages:
① the tremendous amount of details it provides.
② may also be the only way to get certain kinds of information
③ case studies are good ways to study things that are rare.

Disadvantages:
① may not apply to others with similar situations
② observer bias.

The Subjective perspective :-

→ Human behavior is a function of the perceived world, not the objective world.

→ Draws ~~beh~~ from the Gestalt tradition and reacted against the narrowness of behaviorism.

→ we must grasp the person's own 'defination of the situation'.

→ Expected to vary by culture ; personal history, and current motivational state western Vs eastern.

→ Introspective method?

- assume that people fail to see their subjective realities as personal constructions
- Naive realism :- people's tendency to take their constructed, subjective realities to be faithful rendering of an objective world.
- Subjectivist approach also involves systematic observation : Bruner & Goodman, (1947) coin study.

Discriptive method : Survey methods.
[indirect].

- indirect observation through the use of questionnaires
- The only way to find out about very private (covert) behaviour is to ask questions
- Asks a series of questions about the topic they are studing. surveys can be conduted in person in the form of interviews or on the telephone, the internet, or with a questionnaire.

Advantages :-

1. collect private information
2. study large groups of people.

Disadvantages :-

1. choosing a representative sample - studying political opinion.
2. Getting a respresentative sample is not always easy (Banerjee & Chaudhury, 2010)
3. untruthful or distorted responses or forgetting
4. Effect or social desirability
5. courtesy bias.

Experimental method.

- Experiments provide the strongest tests of hypotheses about cause and effect.
- for example, the effects of watching violent cartoon on children

→ What are the steps of the Scientific approach ?

cont'd

→ on children aggressive behaviours.

· Operationalization

variable : is something that can occur with different values.

→ independent variable (IV) : the variable that is manipulated in any experiment

→ Dependent variable (DV) : the response of the participants that is measured and dependent on the IV.

→ If researchers do all of this and find that the children's Behavior is aggressive can they say that the aggressive behavior was caused by the violence in the cartoon ?

→ selection of participants :

→ Deciding on groups :

→ controlling for confounding variables : variables that interfere with each other and their possible effects on some other variable of interest.

→ Genes, length of time, time of day

→ Randomization and control.

→ exp grp

Grp 1 :- those who watch the violent cartoon.

- Grp 2 : those who watch a non violent cartoon. → control grp.

for same length of time.

Result - equally aggressive

conclus : no effect of violent cartoons.

~~no Experiment~~

Experiment Hazard's and controlleng for effects.

* Testing drug on Alzhemimer's patients.
- is it possible that the partiupants who recived the drug knew that they were supposed to improve in memory and, therefore, made a major effor to do so?

* placebo effect : The expectations and biases of the participant in a study can influence their behavior.

placebo : substitute that has no medical effat.

→ ~~Experimenter~~ effect : Even effat animals in a study : aken to observer bias

→ It's possibl that the researcher could give the participants clues about how they are supposed to respond — through the use of body language, tone of voice, or even eye contact

ways to control for these effect.

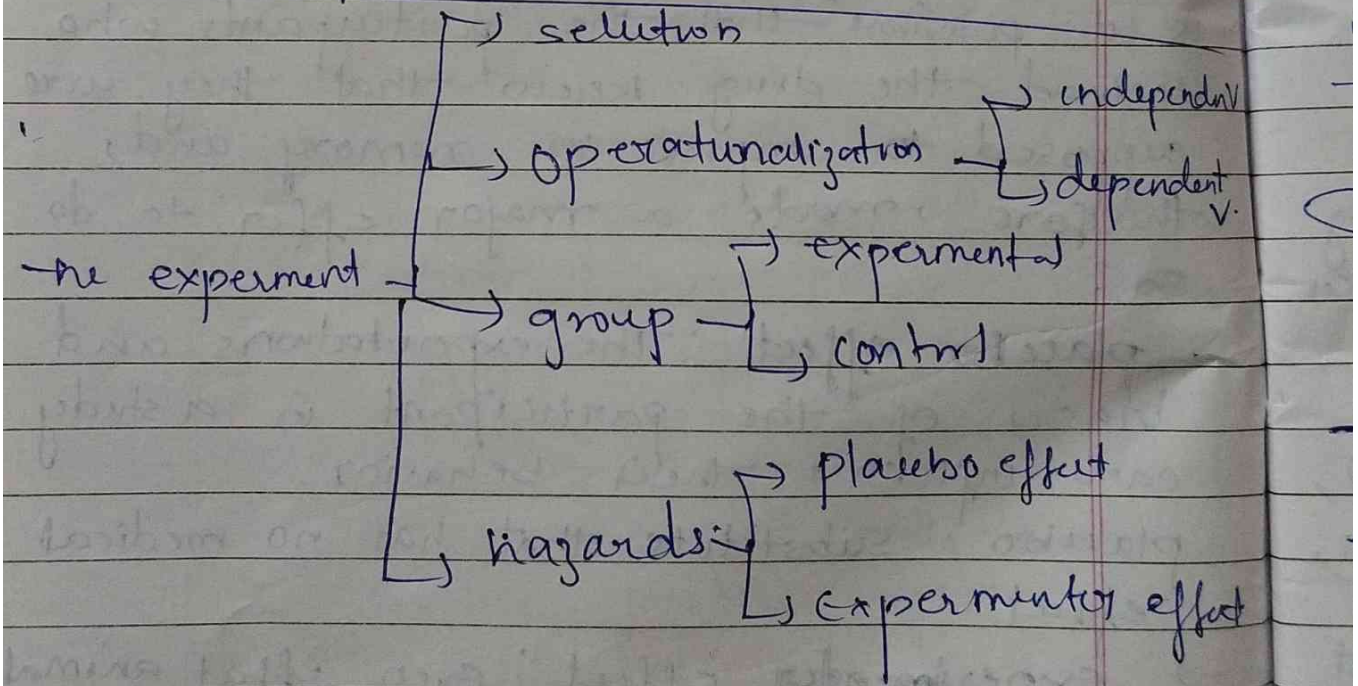
→ singe-blind study.

- study in which the subjects do not know if they are in the experimental or the control group.

→ Double-blind study

- study in which neither the experimenter nor the subjects know if the subjects are in the experimental or the control group.

Single-blind double blind.



Ethics of psychological research.

- Research with humans.

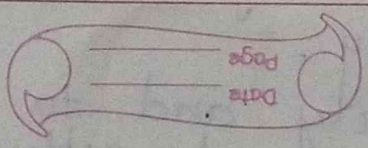
minimal risk: in most cases, the risks anticipated in the research

should be no greater than those ordinarily encountered in daily life.
→ but deciding how much those ~~ordinarily encountered in daily life~~. psychological stress is ethically justified in a research project is not always so clear-cut.

informed consent: participants must be told ahead of time about any aspects of the study that could influence their willingness to cooperate and, after this disclosure, they must enter the study voluntarily and be permitted to withdraw from it at any time they desire without penalty.

But what if the researcher wants to compare participants.

Debriefing: the reasons for keeping them in ignorance - or deceiving them about the procedures must be explained, and any residual emotional reactions must be dealt with so that participants leave with their dignity intact and their appreciation for the research enhanced.



Right to privacy : information about a person acquired during a study must be kept confidential and not made available to others without the research participant's consent.

→ identify through codes
→ report only aggregated data.

2/9/25

Research with animals

* About 7 percent of psychological studies employ animals, 95 percent of which are rats, mice and birds (Atkinson and Hilgard, 2019)
* Advantage : research on animals can produce knowledge that might be impossible or unethical to obtain from humans.
* Animal research has in fact played a pivotal role in understanding and treating psychological problems such as anxiety, stress aggression, depression, drug abuse, eating disorders, hypertension, and Alzheimer's drug abuse, eating disorders, hypertension and Alzheimer's disease (Carroll & Overmier 2001).
* Federal and APA guidelines, require that any painful or harmful procedures impose on animals must be thoroughly justified in terms of the knowledge to be gained from the study.
* Treat animals humanely and to minimize their pain and suffering.

UNIT - 2

- EMOTIONS

TED - BUNDY - Serial killer - killed women for pleasure.

→ Psychopath and dead people doesn't show emotions.

→ Definations

→ Latin root word mot, meaning "to move."

→ The "feeling" aspect of consciousness; characterized by three elements:
a certain physical arousal,
a certain behavior that reveals the feeling to the outside world, and
an inner awareness of the feeling

→ An emotion is a complex, multicomponent episode that creates.

The physiology of emotions

→ Arousal is created by the sympathetic nervous system

→ The heart rate increases.
breathing becomes more rapid,
the pupils dilate,
and the mouth may become dry
Anger and fear.

→ Although facial expressions do differ among various emotional responses.

Lab exp

→ sadness, anger, fear are associated with greater increase in heart rate than is disgust.

→ higher increases in skin conductance occur during disgust as compared to happiness.

→ anger is more often associated with vascular measures, such as higher ~~diatolic~~ bl. diastolic blood pressure, as compared to fear.

The polygraph test

→ the idea was that lying would produce different physiological reactions from telling the truth (Bell & Gruber)

→ This assumption has proved to be false.

→ improvements: concealed information test (or CIT)

→ useful tool to convince suspects to confess to their crimes: it

→ really only detects the physiological correlates of emotion in general, not lying in specific.

The Behavior of Emotion: Emotional Expressions.

→ anger
→ sadness → fear.
→ disgust → surprise.
→ happiness → contempt.

→ Charles Darwin (1998) : facial muscles evolved to communicate specific information to onlookers.
→ not in line with the behaviorist movement of the early and middle twentieth century.
→ universal nature to at least seven basic emotions
→ children blind from birth can produce the appropriate facial expressions.

* contempt is the only unilateral expression
* It can occur with or without a hint of a smile or angry expression.

. DISPLAY RULES.
→ learned ways of controlling displays of emotion in social settings.
→ Exactly when, where, and how an emotion is expressed may be determined by the culture.
→ vary from culture to culture.
→ learned ways of controlling displays of emotion in social settings.
→ exp - Japanese people have strict social rules about showing emotion in public situations - they simply do not show emotion, remaining cool, calm, and collected, at least on the outside.

* different between individualistic and collectivistic cultures : USA vs Japan.

* different for males and females.

→ boys are reluctant to talk about feelings in a social setting.

→ adults, researchers looking

Subjective Experience : labeling Emotion

→ interpreting the subjective feeling by giving it a label : anger, fear, disgust happiness, sadness, shame, interest, etc.

→ "cognitive element"

• LEBEL : is at least in part a learned response influenced by their language and culture.

→ Chinese Americans who were still firmly rooted in their original Chinese culture - use labels that refered to bodily sensations. such as "dizzy" or social relationship such as "friendship" Vs

→ "Americanized" Chinese Americans and European Americans, who tended to use more directly emotional words (such as "sad or love")

· Kitayama

Emotions Vs moods.

① Emotions typically have a clear cause they are about something or someone:
- you are angry at your sis, you are awestruck by grand canyon

② moods on other hand, are often free-floating and diffuse affective states. for unknown reasons, you feel irritable one day, and cheerful the next.

② Emotions are typically brief, lasting only seconds or minutes; but moods endure longer, lasting for hours, even days.

③ Emotions typically implicate the multiple component systems described.

Emotions in Indian tradition.

→ theory of "rasa"

- Rasa: refined subjective mental state.
- refined to an inherent quality that is cherished and constitutes the emotive content of a piece of art.

→ The person as a dancer / poet and spectator / audience is transported to an experienced state which may not be congruent with the mundane experience.

→ Bhava: An emotion.

Sage Bharata synthesised the rasa theory in the context of theatre - later extended to all poetry.

→ Natya Shastra.

→ Rasa exists in vedic hymns: considered a stream of consciousness

→ is abstract: hence universal across literary criticism dance and visual arts.

→ wonder, happy, peace, compassion, love, music.

Early theories of Emotion.

The common-sense theory of Emotion.

common sense theory	Stimuli	First response	Second response
"I'm shaking bcaus im afraid.	Snarling dog	Concious fear.	ANS arousal.

→ in the common-sense theory of emotion, a stimulus (snarling dog) leads to an emotion of fear.

James-lange theory of Emotion

	Stimuli	first response	second response
James-lange theory	snarling dog	ANS arousal, changes in body → Autonomic nervous system.	conscious fear.

If this is true

→ what about people who have spinal cord injuries that prevent the sympathetic nervous system from functioning?

→ not really, several studies of people with spinal cord injuries report that these people are capable.

Cannon-Bard theory of Emotion

	Stimuli	first response	second response
cannon Bard theory	snarling dog.	subcortical brain activity	ANS arousal changes in body ↓ conscious fear.

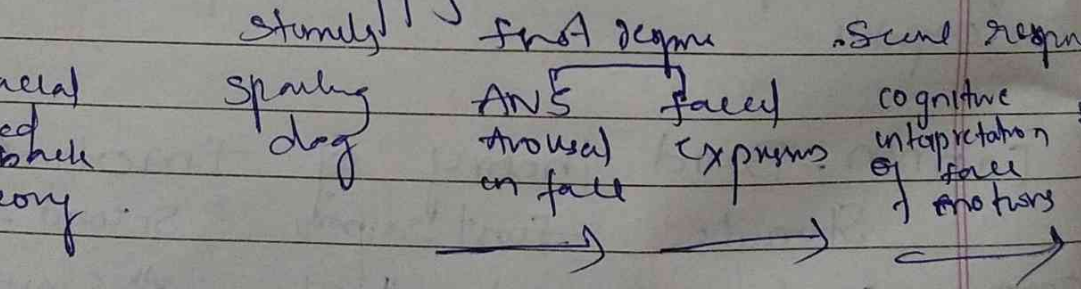
Criticism of canon - Bard theory.

Lashley started that the thalamus would have to be pretty sophisticated to make sense of all the possible human emotions and relay them to the proper areas of the cortex and body.

- people with spinal cord injuries, which appear to suggest that emotions can be experienced without feedback from the sympathetic organs to the cortex.

- other areas.

The facial feedback Hypothesis: smile you'll feel happy.



Cognitive theories of emotion.

Schachter - Singer cognitive Arousal theory of emotion.

→ Schachter and singer

→ Two-factor theory.

→ two things have to happen before emotion occurs: the physical arousal and a labeling of the arousal based on cues from the surrounding environment.

- stimulus → first response → cognitive appraisal → ANS arousal changes in body → second response → fear

Snarling dog

I am aroused in the presence of scary dog; therefore, I must be afraid.

Misattribution of arousal.

- less limited effect than the two factor theory.

- lingering physiological arousal - say, from running up four flights of stairs - can be mistakenly attributed to subsequent circumstances like an ambiguous remark, intensify

lazarus and the cognitive - mediational theory of Emotion.

the snarling dog is dangerous and therefore I should feel afraid

stimulus Barking dog

first response Appraisal (fear) of threat

Sound

Body response

Criticisms

- Some researches belive that emotional reactions to situations are so fast that they are almost instantaneous, which would have little time for a cognitive to occur first

→ Others have found that the human brain can respond to a physical threat before conscious thought enters the picture

→ the amygdala can prompt emotional

Which theory is right.

- All of the theories are correct to at least some degrees.
- falling in love
- dangerous situation.

Tought and action tendencies and emotion.

~~Emotion~~ thought - action tendencies: the way feeling guide behavior and information processing is through the urges that accompany them.

Emotions are their associated though. action tendencies.

Emotion	thought action tendency
Anger	Attack
fear	Escape
Disgust	Expel
Guilt	make amends
shame	Disappear
sadness	withdraw
joy	play
interest	explore.
contentment	savor and integrate
pride	Dream big
Gratitude	Be prosocial.
Elevation	Became a better person.

whether urges become actions depends on the complex interplay of impulse control, cultural norms, and other factors.

physiology of emotions: BODILY CHANGES AND EMOTIONS:

- result from activation of the sympathetic division of the autonomic nervous system

- Sympathetic nervous system prepares the body for emergency action and is responsible for the following changes

(1) Blood pressure and heart rate increase
(2) Respiration becomes more rapid.
(3) The pupils dilate
(4) perspiration increases while secretion of salva and mucus decreases.
(5) Blood-sugar level increases to provide more energy.
(6) the blood clots more quickly in case of wounds
(7) Blood is diverted from the stomach and intestines to the brain and skeletal muscles.
(8) The hairs on the skin become erect, causing goose pimples.

Sympathetic nervous system thus gears up the organms to espend energy

Apprausals in the brian.

Amygdala ——→ part of brain

Rolls of Amygdala.

Actuks :-

undoing effect of positive emotions.